ACHELION CAPITAL

ARMS TECHNICAL HANDOFF BRIEF

System Architecture, Interface Contracts & Integration Pipeline

v4.0 · March 2026 · Architecture AB · 58/20/14/8 · Seven-Pillar + Seven-Gap Upgrades

VERSION	KEY CHANGES
v3.1 → v4.0	Seven world-class risk upgrades added: (1) Factor Exposure Monitor — FEM, (2) Volatility-Adjusted Re-entry Sizing — VARES, (3) Portfolio Drawdown Sentinel — PDS, (4) Liquidity-Adjusted Execution Protocol — LAEP, (5) Permanent Tail Risk Hedge — PTRH using QQQ puts, (6) Conviction Decay Function — CDF, (7) Stress Scenario Library — SSL. Four new dataclasses. Four new database tables. New file: tail_hedge.py, stress_scenarios.py, drawdown_sentinel.py, factor_exposure.py.
v3.1 base	All v3.1 features retained: Adaptive signal weighting · Regime Confidence Score · Escalation Rule v2 (cumulative 2.5% + auto-suppress) · ARES Gate 4 · Conviction Drift Monitor · Tiered SLOF · LP Auto Narrative · Pre-Staging SYS-3B · Contagion Velocity SYS-5B · Nowcast/Forecast Divergence · Backtesting Infrastructure · Fund-agnostic Config Layer · Data Feed Resilience · Multi-portfolio Governance · Model Validation Requirements · Transaction Cost Model · Version Governance.
STATUS	CONFIDENTIAL — Development team and authorized GP use only. v4.0 is the canonical build target for Josh. All prior versions superseded.

1. AUTHORITY HIERARCHY — UNCHANGED IN V4.0

The seven-pillar authority hierarchy is unchanged. New v4.0 modules are ADVISORY or ADDITIVE — they tighten ceilings further or provide information, but they never loosen what ARAS has set. The Portfolio Drawdown Sentinel (PDS) is the only new module with ceiling authority, and its ceiling is always at or below ARAS.

AUTHORITY	RULE
ARAS — Highest	Absolute exposure ceiling. Cannot be exceeded by any module, human, or override.
PDS — New v4.0	Portfolio Drawdown Sentinel sets its own ceiling. LOWER of ARAS or PDS ceiling always prevails. PDS can only make the portfolio MORE defensive than ARAS, never less.
FEM — Advisory	Factor Exposure Monitor is ADVISORY. It flags concentration risk to the PM but does not override ARAS or PDS ceilings.
PTRH — Additive	Permanent Tail Risk Hedge exists OUTSIDE the exposure ceiling calculation. QQQ puts are protective instruments, not offensive exposure.

2. GAP 1 — FACTOR EXPOSURE MONITOR (FEM)

The FEM is a new ARAS sub-module that detects hidden concentration risk across positions. ARMS can be in RISK_ON at 100% ceiling while effectively running a single concentrated bet — AI semiconductor, Taiwan manufacturing, BTC beta, or dollar sensitivity. The FEM surfaces this to the PM before it becomes a drawdown.

2.1 Factor Definitions

FACTOR	POSITIONS AFFECTED	WEIGHT METHOD
AI Capex Cycle	NVDA, AMD, ALAB, MU, MRVL, AVGO, ANET, ARM	Beta-weighted position size × AI revenue exposure %
Taiwan Manufacturing	NVDA, AMD, ALAB, MU, MRVL, ARM (TSMC-dependent)	Sum of weights of TSMC-dependent positions
BTC Beta	IBIT, ETHB, BSOL, PLTR (BTC treasury), TSLA (BTC treasury)	BTC correlation coefficient × position weight

FACTOR	POSITIONS AFFECTED	WEIGHT METHOD
Dollar Sensitivity	All equity + BTC (DXY inverse correlation)	90-day DXY correlation × position weight (signed)
Rate Sensitivity	VRT, ETN, ANET (infrastructure capex), SGOV (positive)	Duration-equivalent sensitivity to 100bps rate move

2.2 Thresholds and Actions

LEVEL	THRESHOLD	ACTION
WATCH	>65% single factor	PM alert on daily monitor. No mandatory action. Document in session log.
ALERT	>80% single factor	Mandatory PM review within 24 hours. If not resolved by conviction update or trim, escalate to ARAS as input signal (advisory +0.05 composite).
Cross-sleeve CORR	>0.70 equity/crypto 30d	Advisory flag. Feeds existing SYS-5 / SYS-5B contagion monitors. Not a new ceiling.

2.3 Interface Contract — FactorExposureSignal

```
@dataclass class FactorExposure:     factor_name: str          # e.g., 'AI_CAPEX_CYCLE'
exposure_pct: float       # 0.0-1.0 of portfolio NAV     positions_affected: List[str]
threshold_watch: float    # 0.65     threshold_alert: float    # 0.80     status:
Literal['NORMAL','WATCH','ALERT']  @dataclass class FactorExposureSignal:     as_of: str
factors: List[FactorExposure]     highest_exposure_factor: str     highest_exposure_pct: float
cross_sleeve_correlation: float    # rolling 30d equity/crypto     advisory_composite_add: float
# 0.0 normally, +0.05 if ALERT
```

2.4 File: src/modules/factor_exposure.py

Runs daily after portfolio snapshot is calculated. Inputs: position weights, beta estimates, correlation matrix (rolling 90d). Output: FactorExposureSignal. Feeds into ARAS composite as advisory input only. Feeds into daily monitor display.

3. GAP 2 — VOLATILITY-ADJUSTED RE-ENTRY SIZING (VARES)

ARES re-entry is enhanced with volatility-adjusted tranche sizing. Re-entry tranche size is inversely proportional to current VIX relative to its 90-day moving average. When volatility is elevated at re-entry, position sizes are smaller. When volatility is normalized, position sizes return to full. This is standard practice at every major risk-managed fund.

3.1 VARES Formula

```
# Vol adjustment factor vix_90d_avg = mean(VIX, lookback=90) vol_adj_factor = min(1.0,
vix_90d_avg / vix_current)  # Applied to base tranche size base_tranche_pct = 0.33   # always 3
equal tranches of target adjusted_tranche = base_tranche_pct * vol_adj_factor  # Hard limits
adjusted_tranche = max(0.15, min(0.35, adjusted_tranche))  # Example: # VIX current = 35, VIX 90d
avg = 20 # vol_adj_factor = 20/35 = 0.571 # adjusted_tranche = 0.33 * 0.571 = 0.188 (18.8% of
target per tranche) # Result: you come back at 57% of normal tranche size when vol is elevated
```

3.2 Tranche Timing

TRANCHE	TIMING	SIZE	CONDITION
T1	Gate 4 clear + PM approval	**base × vol_adj_factor**	ARES gates 1-4 all clear. Composite below DEFENSIVE threshold.
T2	T1 + 48 hours	**Same as T1**	Regime held for 48h post-T1. No new circuit breaker. Composite stable or declining.

TRANCHE	TIMING	SIZE	CONDITION
T3	T2 + 48 hours	**Remainder to target**	Regime held for 48h post-T2. Full deployment reached. SLOF reactivation PM discretion.
ABORT	Any tranche window	**Halt all tranches**	If any circuit breaker fires between tranches: abort re-entry, return to current regime ceiling.

3.3 Extended ReentryPlan Dataclass

```
@dataclass class ReentryPlan:                    # extends existing ARES     as_of: str
current_regime: str     target_exposure: float     vix_current: float     vix_90d_avg: float
vol_adj_factor: float             # NEW     tranche_size_pct: float           # NEW — adjusted
tranche     tranche_1_target: float           # NEW     tranche_2_target: float           # NEW
tranche_3_target: float           # NEW     tranche_1_executed: bool     tranche_2_due:
Optional[str]      # ISO datetime     tranche_3_due: Optional[str]     abort_triggered: bool
# NEW
```

4. GAP 3 — PORTFOLIO DRAWDOWN SENTINEL (PDS)

CRITICAL — INDEPENDENT MODULE

The Portfolio Drawdown Sentinel operates INDEPENDENTLY of ARAS. It monitors portfolio NAV against its rolling high-water mark. It is the last line of defense when ARAS is wrong or slow. PDS and ARAS ceilings are always compared — the lower ceiling always prevails. ARAS cannot override PDS. PDS cannot override ARAS. They coexist, each setting a floor on risk.

4.1 Drawdown Thresholds

LEVEL	DRAWDOWN FROM HWM	ACTION	CEILING
WARNING	−8% from HWM	PM alert. No mandatory action. Heightened monitoring. Document in daily log.	None — advisory only
DELEVERAGE 1	−12% from HWM	Automatic — force portfolio to 60% gross ceiling. Independent of ARAS regime.	**60% gross**
DELEVERAGE 2	−18% from HWM	Automatic — force portfolio to 30% gross ceiling. PM notified. LP communication triggered.	**30% gross**

4.2 High-Water Mark Rules

The HWM is the portfolio's highest ever closing NAV. It updates daily when a new high is set. It never decrements — it only moves up. When the portfolio closes below the prior HWM, the drawdown clock starts. The drawdown is calculated as (current_NAV / HWM) – 1.

4.3 PDS + ARAS Interaction Rule

```
# Each session, determine effective ceiling: aras_ceiling = aras_output.exposure_ceiling        #
e.g., 1.00 (RISK_ON) pds_ceiling  = drawdown_sentinel.ceiling           # e.g., 0.60 (DELEVERAGE
1) effective_ceiling = min(aras_ceiling, pds_ceiling) # 0.60 prevails  # PDS ceiling persists
until: # 1. Portfolio NAV recovers above HWM - 8% (exit WARNING) # 2. Portfolio NAV recovers
above HWM - 12% (exit DELEVERAGE 1) # 3. Explicit PM override with documented justification (PDS
only — ARAS cannot be overridden)
```

4.4 DrawdownSentinelSignal Dataclass

```
@dataclass class DrawdownSentinelSignal:     as_of: str     current_nav: float
high_water_mark: float     drawdown_pct: float               # negative number     status:
```

```
Literal['NORMAL','WARNING','DELEVERAGE_1','DELEVERAGE_2']     pds_ceiling: float                #
1.0 if NORMAL, 0.60 or 0.30 otherwise     effective_ceiling: float          # min(aras_ceiling,
pds_ceiling)     hwm_date: str                     # date HWM was set     days_since_hwm: int
```

File: src/engine/drawdown_sentinel.py. Runs daily after NAV calculation. DB table: drawdown_sentinel_log. Feeds into daily monitor as a separate status banner.

5. GAP 4 — LIQUIDITY-ADJUSTED EXECUTION PROTOCOL (LAEP)

SYS-3 execution methodology adjusts automatically based on market liquidity conditions at the time of execution. In normal markets, market orders execute within 30 minutes. In elevated volatility, VWAP orders extend to 60 minutes. In crisis, VWAP extends to 90 minutes with priority reordering to protect the least liquid positions.

5.1 Liquidity Mode Classification

MODE	VIX TRIGGER	ORDER TYPE	WINDOW	SLIPPAGE BUDGET	PRIORITY ORDER
NORMAL	VIX < 25	Market	30 min	8 bps est.	P1→P2→P3→P4
ELEVATED	VIX 25–45	VWAP	60 min	20 bps est.	P1→P2→P3→P4
CRISIS	VIX > 45	VWAP	90 min	40 bps est.	BSOL→ETHB→IBIT→equity

CRISIS PRIORITY REORDER RATIONALE

In CRISIS mode (VIX >45), BSOL is executed first because it is the least liquid crypto ETF — the widest spread, highest market impact. ETHB second (newer, less liquid than IBIT). IBIT third (most liquid crypto ETF). Equity last because major ETF components (TSLA, NVDA) have the deepest liquidity even in crisis. This is the opposite of normal priority — in crisis, you exit the illiquid positions first while you still can.

5.2 LiquidityMode Enum + Extended OrderBookEntry

```
class LiquidityMode(Enum):     NORMAL   = "NORMAL"     # VIX < 25     ELEVATED = "ELEVATED"   #
VIX 25-45     CRISIS   = "CRISIS"     # VIX > 45  @dataclass class OrderBookEntry:              #
extends existing     ticker: str     action: Literal['SELL','BUY','CLOSE']     shares: int
dollar_amount: float     priority: int     reason: str     liquidity_mode: LiquidityMode  # NEW
order_type: str                # NEW: 'MARKET' | 'VWAP'     execution_window_min: int      # NEW:
30 | 60 | 90     slippage_budget_bps: float     # NEW: 8 | 20 | 40
```

6. GAP 5 — PERMANENT TAIL RISK HEDGE (PTRH)

QQQ PUTS — NOT SPX

The PTRH uses QQQ puts, not SPX puts. Rationale: Achelion's 58% equity sleeve is concentrated in AI infrastructure names that trade as NASDAQ/QQQ proxies. In a technology-specific drawdown — the primary risk — QQQ falls materially harder than SPX (2022: QQQ −32.6% vs SPX −18.1%). QQQ puts provide approximately 1.8x the protection of SPX puts per dollar of premium in the scenarios most relevant to this portfolio.

6.1 Hedge Structure

PARAMETER	SPECIFICATION
Instrument	QQQ put options — listed, exchange-traded
Strike	10–15% out-of-the-money from current QQQ price. Recalibrate at each roll.
Expiry	60–90 day options. Roll at 30 DTE to avoid gamma decay acceleration.

PARAMETER	SPECIFICATION
Target protection	10–15% portfolio protection in a −30% QQQ scenario. Size to achieve this payoff profile.
Annual premium cost	Target 0.30–0.50% NAV per year at base size. At $50M AUM: approx $150K–$250K/year.
Relationship to SLOF	PTRH is a protective put position — it does NOT count against SLOF overlay limits. It is a separate sleeve in the options budget.
Exposure ceiling	**PTRH is EXEMPT from ARAS gross exposure calculation — identical treatment to the defensive sleeve.**

6.2 Regime-Dependent Sizing

ADDENDUM 4 SUPERSESSION NOTICE (March 27, 2026): The CRASH row in the table below has been updated. The prior specification ("Hold at 150% — no new buys") is superseded by the Coverage Adequacy Model (CAM) specified in Addendum 4 to FSD v1.1. In CRASH regime, CAM autonomously calculates required PTRH coverage every 5 minutes using four multipliers (regime score, equity exposure, FEM concentration, macro stress) and auto-corrects notional to the required level. The 2.0× minimum in CRASH is a floor — CAM may require higher coverage depending on current risk configuration. The PM is never asked about PTRH sizing under any conditions after CAM is live. See src/engine/cam.py.

REGIME	PTRH SIZE	STRC DYNAMIC LAYER	COMBINED PROTECTION
RISK_ON	100% base (1.0×)	Locked — not deployed	Permanent hedge only
WATCH	125% base (1.25×)	Locked	Permanent + 25% upsize
NEUTRAL	125% base	50% reserve deployed (QQQ puts)	Permanent + STRC reserve layer
DEFENSIVE	150% base (1.5×)	100% reserve deployed	**Maximum hedge stack**
CRASH	2.0× base (CAM autonomous) — supersedes prior "hold at 150%" rule. See Addendum 4.	All in	CAM calculates required coverage every 5 min. Auto-corrects notional. PM not involved. Addendum 4 governs.

6.3 TailHedgeStatus Dataclass

```
@dataclass class TailHedgeStatus:     as_of: str     current_regime: str     base_size_notional: float          # target notional protection     current_size_multiplier: float     # 1.0 | 1.25 | 1.50 | 2.0 (CRASH via CAM — Addendum 4)     current_positions: List[OptionsPosition] next_roll_date: str                # when 30 DTE is reached     estimated_annual_premium_drag: float   # % NAV     payoff_at_minus30pct_qqq: float    # estimated $ protection     db_table = "tail_hedge_positions"  @dataclass class OptionsPosition:     ticker: str # "QQQ"     option_type: Literal['PUT']     strike: float     expiry: str     contracts: int premium_paid: float     current_value: float
```

7. GAP 6 — CONVICTION DECAY FUNCTION (CDF)

The CDF extends the existing Conviction Drift Monitor to make it actionable. The Drift Monitor currently flags positions that deviate >20% from their C²-implied contribution after 45 days. The CDF adds automatic conviction weight reduction when a position underperforms its thesis — preventing the most dangerous behavior in concentrated portfolios: doubling down on a broken thesis.

7.1 Underperformance Trigger

```
# CDF fires when BOTH conditions are met for N consecutive days: condition_1 = position_return_45d < (qqq_return_45d - 0.10)  # -10pp vs QQQ condition_2 = days_underperforming >= 45  # Decay schedule if days_underperforming >= 45:     conviction_decay_factor = 0.80   # C²-weight × 0.80 (C10 → effective C9) if days_underperforming >= 90:     conviction_decay_factor = 0.60   # C²-weight × 0.60 (C10 → effective C8) if days_underperforming >= 135:
```

```
conviction_decay_factor = 0.60   # Hold at 0.60 — mandatory PM review  # Reset condition if
position_outperforms_qqq_by_5pp_over_15d:     conviction_decay_factor = 1.00   # Full restoration
```

7.2 Impact on Position Sizing

The conviction_decay_factor multiplies the C^2 weight in the Master Engine. A position with C10 (C^2=100) and decay_factor=0.80 behaves as if it had C^2=80 (C9) for position sizing purposes. This means the position is sized smaller without requiring an explicit PM sell decision. The PM retains full control — they can override the decay with a documented conviction reaffirmation.

WHAT CDF DOES NOT DO

CDF does not force a sell. It does not trigger a regime change. It does not reduce the SLOF eligibility of the position. It only reduces the position's effective weight in the Master Engine portfolio construction calculation. The PM can override at any time with a documented justification. The decay is visible on the daily monitor as a flag next to the position.

7.3 PortfolioTarget Extension

```
# Extend existing PortfolioTarget dataclass @dataclass class PositionTarget:     ticker: str
conviction_level: int          # original 1-10     conviction_decay_factor: float  # NEW: 1.0
normal, 0.80/0.60 decayed     effective_c_squared: float      # NEW: conviction^2 * decay_factor
target_weight: float     decay_days: int                 # NEW: 0 if no decay
decay_triggered: bool           # NEW
```

8. GAP 7 — STRESS SCENARIO LIBRARY (SSL)

The SSL runs daily after portfolio snapshot. It applies eight defined historical and custom stress scenarios to the current portfolio weights and outputs estimated P&L for each. This gives the PM information the regime model cannot provide — not what has already happened, but what would happen if a defined stress event occurred today.

8.1 Scenario Definitions

ID	NAME	EQUITY (QQQ)	BTC	DEFENSE	SOURCE
S1	2008 Lehman Collapse	**−35%**	N/A	Agg +5%, Gold +12%	Sep–Nov 2008 actual returns
S2	2020 COVID Crash	**−34%**	**−53%**	Agg +3%	Feb 20 – Mar 23, 2020
S3	2022 Rate Shock	**−22%**	**−35%**	Agg −5%, DBMF +8%	Q1 2022 actual
S4	Flash Crash — Intraday	**−10% in 20 min**	**−15%**	Flat	Custom — worst-case single session
S5	Hormuz Closed 90 Days	**−15%**	**−10%**	SGOL +18%, DBMF +12%	Custom — current conflict extended
S6	Crypto Bear — BTC −60%	−5%	**−60% BTC, −65% ETH**	Flat	2022 crypto collapse analog
S7	Fed Shock +100bps	**−12%**	**−20%**	SGOV +1%, Agg −8%	Custom — surprise tightening
S8	AI Bubble Unwind −40%	**−40% (AI names only)**	**−20%**	SGOL +8%	Custom — thesis-specific stress

8.2 ScenarioResults Dataclass

```
@dataclass class ScenarioPnL:     scenario_id: str              # e.g., 'S1'     scenario_name:
str     portfolio_pnl_pct: float      # estimated % loss     portfolio_pnl_usd: float      #
estimated $ loss at current NAV     largest_position_loss: str    # "NVDA: -$X.XM"
hedge_offset_usd: float       # PTRH estimated payoff in this scenario     net_pnl_after_hedge:
```

```
float    # portfolio_pnl_usd + hedge_offset_usd  @dataclass class ScenarioResults:     as_of: str
current_nav: float     scenarios: List[ScenarioPnL]     worst_scenario: str           #
scenario_id     worst_net_loss_pct: float     # after hedge     db_table = "scenario_pnl_log"
```

File: src/modules/stress_scenarios.py. Computation is a simple matrix multiplication: scenario_returns × current_position_weights = scenario_pnl. Runs in milliseconds. No live data required beyond the daily portfolio snapshot.

9. UPDATED DATABASE SCHEMA — V4.0 ADDITIONS

NEW TABLE	PURPOSE	KEY COLUMNS
factor_exposure_log	Daily factor concentration snapshots	date, factor_name, exposure_pct, status, positions_affected
drawdown_sentinel_log	NAV vs HWM tracking and PDS status	date, nav, hwm, drawdown_pct, status, pds_ceiling, effective_ceiling
tail_hedge_positions	Current QQQ put positions and PTRH status	date, strike, expiry, contracts, premium_paid, current_value, size_multiplier, regime
scenario_pnl_log	Daily scenario library output	date, scenario_id, portfolio_pnl_pct, portfolio_pnl_usd, hedge_offset, net_pnl

10. UPDATED FILE STRUCTURE — V4.0

```
achelion-arms/ ├── src/ │   ├── engine/ │   │   ├── aras.py                    # unchanged core │
│   ├── macro_compass.py           # unchanged │   │   ├── master_engine.py           # updated:
conviction_decay_factor input │   │   ├── kevlar.py                  # unchanged │   │   ├──
perm.py                     # UPDATED: CDF — conviction decay function │   │   ├── slof.py
# unchanged │   │   ├── ares.py                    # UPDATED: VARES — vol-adjusted tranche sizing
│   │   ├── drawdown_sentinel.py       # NEW: Portfolio Drawdown Sentinel │   │   └──
tail_hedge.py               # NEW: Permanent Tail Risk Hedge │   ├── modules/
# ARAS sub-modules │   │   ├── deleveraging_risk.py       # unchanged │   │   ├──
crypto_microstructure.py    # unchanged │   │   ├── margin_stress.py           # unchanged │   │
├── dealer_gamma.py             # unchanged │   │   ├── pcr_regime.py              # unchanged │
│   ├── shutdown_risk.py           # unchanged │   │   ├── factor_exposure.py         # NEW:
Factor Exposure Monitor │   │   └── stress_scenarios.py        # NEW: Stress Scenario Library │
├── execution/ │   │   ├── circuit_breaker.py         # unchanged │   │   ├── order_book.py
# UPDATED: LAEP — liquidity-adjusted execution │   │   ├── overnight_monitor.py       # unchanged
│   │   ├── correlation_monitor.py     # unchanged │   │   ├── escalation_engine.py       #
unchanged │   │   ├── trade_order_generator.py   # unchanged │   │   └── pm_protocol.py
# unchanged │   ├── data_feeds/                    # unchanged │   └── api/ │       └──
dashboard_api.py            # UPDATED: new module outputs in daily report
```

11. DAILY ASSESSMENT — V4.0 ADDITIONS

The daily PM assessment output adds three new sections from the v4.0 upgrades. These appear in the daily monitor after the existing ARAS composite and circuit breaker sections.

NEW SECTION	WHAT IT SHOWS	ACTION TRIGGER
Factor Exposure Status	Concentration by factor (AI Capex, Taiwan, BTC Beta, etc.) with status badges	ALERT badge → mandatory PM review within 24h
Drawdown Sentinel	Current NAV vs HWM, drawdown %, PDS status, effective ceiling	DELEVERAGE_1 or DELEVERAGE_2 → automatic execution
Tail Hedge Status	Current QQQ put positions, size multiplier, next roll date, annual drag	Roll required when DTE ≤ 30

NEW SECTION	WHAT IT SHOWS	ACTION TRIGGER
Stress Scenario P&L	Table: all 8 scenarios × current portfolio. Gross and net-of-hedge P&L.	Advisory — worst scenario shown prominently
Conviction Decay Flags	Positions with active decay factor — decay level, days decaying, reset condition	Day 135 → mandatory fundamental review

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CONFIDENTIAL: This document is confidential and intended solely for authorized development personnel and Achelion Capital GPs. v4.0 represents the complete world-class risk management specification. All seven institutional gaps identified in the strategic review have been closed. Architecture AB v4.0 · March 2026.